Regular polytopes with projective special linear groups

Human Compression project

For projective groups, the common theory [U] classifies string markings with two proper end groups, without assuming the intersection property. Here we apply that theorem to the classification of Leemans and Schulte [LS]. All group and polytope isomorphisms preserve the ordered distinguished generators.

Theorem 1 (Leemans–Schulte)**.** *Let q be a prime power. A regular abstract polytope of rank at least four with full automorphism group $\mathrm{PSL}_2(q)$ has rank four and belongs to exactly one of the following two classes:*

q	*Polytope*	*Type*	*Face vector*
11	*11-cell*	$\{3,5,3\}$	$(11,55,55,11)$
19	*57-cell*	$\{5,3,5\}$	$(57,171,171,57)$

The 11-cell has hemi-icosahedral facets and hemi-dodecahedral vertex figures; these local maps are interchanged for the 57-cell. Both polytopes are self-dual and universal for their stated facets and vertex figures. The edge graph of the 11-cell is K_{11}. In both cases the even-word subgroup is the full automorphism group.

Proof. Let $(r_0, \dots, r_{n-1})$ be the distinguished string C-marking. The intersection property makes every subgroup generated by a proper subset of the marking proper: if r_i belonged to the subgroup generated by the other generators, their intersection with $\langle r_i \rangle$ would be both trivial and nontrivial. In particular both end groups are proper.

The projective restriction and finite classification theorems of [U] therefore give odd q, rank four, and precisely the marked groups $\mathrm{PSL}_2(11)$, $\mathrm{PSL}_2(19)$ and S_5. The small group $\mathrm{PSL}_2(3) \cong A_4$ is excluded because its involutions generate only the normal four-group. For the remaining odd q, the group $\mathrm{PSL}_2(q)$ is simple and hence cannot be S_5. Its order $q(q^2-1)/2$ is strictly increasing for positive q, so the other two groups force $q = 11$ and $q = 19$ respectively.

Conversely, [U] proves the intersection property and uniqueness up to an isomorphism preserving the ordered generators for each surviving presentation. The group–polytope correspondence gives exactly the stated regular polytopes and their full automorphism groups. The geometry theorem of [U] supplies their local maps, face vectors, reversal automorphisms, even-word subgroups and universal covering property. It also proves the edge incidence statement, rather than inferring it from the face vector alone. □

This recovers [LS, Theorems 1, 3 and 4], the introductory rank restriction, and the stated geometric conclusions of its Section 5. The complete table of ten presentations in [LS] is proved and printed once in [U]. The 11-cell and 57-cell retain their attribution to Grünbaum and Coxeter; the classification above is due to Leemans and Schulte. Historical subgroup multiplicity formulas and discussion of other presentations and chiral constructions are outside the results recovered here.

The full proofs and finite program of [U] are required dependencies. The common paper and program are counted once in the measurement for this topic. Dickson's classification and the classical group–polytope and covering correspondences remain explicit inputs there. The available Lean certificates cover stated supporting lemmas; they do not formalize this complete classification.

References

[U] *Common theory for projective string markings*, companion paper, projective restriction, finite classification and geometry theorems.

[LS] D. Leemans and E. Schulte, *Groups of type $L_2(q)$ acting on polytopes*, arXiv:math/0606660v1, the bound fourteen-page author edition; *Advances in Geometry* **7** (2007), 529–539.